

■ SignBridge AI

Official Sign Language Comprehensive Handbook & Visual Guide

Authoritative Reference Manual for 500+ Signs, Anatomical Formations, Motions & Camera AI Protocol

Handbook Purpose: This official guide documents standardized manual signs for education, healthcare, emergency triage, daily communication, and assistive artificial intelligence systems. Each sign contains the exact physical handshape, starting body position, motion trajectory, and practical conversational context.

■ Section 1: Standard Fingerspelling Alphabet (A–Z)

Standard ASL / ISL manual alphabet shapes used for spelling names, acronyms, and specialized terminology.

■ Section 2: Greetings (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
HELLO [Beginner]	Open 5 fingers spread wide facing camera.	Gentle outward wave from temple.	Standard greeting to begin a conversation.
GOOD MORNING [Beginner]	Sign 'Good' followed by sun rising arm motion.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOOD MORNING'.
GOOD AFTERNOON [Beginner]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
GOOD EVENING [Beginner]	Sign 'Good' followed by sunset wrist downward tap.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOOD EVENING'.
GOOD NIGHT [Beginner]	Sign 'Good' then cup hands over horizontal forearm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOOD NIGHT'.
WELCOME [Beginner]	Open palms scooped inwards towards chest.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WELCOME'.
NICE TO MEET YOU [Beginner]	Wipe palms horizontally, then index fingers meet.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NICE TO MEET YOU'.
HOW ARE YOU [Beginner]	Curved knuckles roll outward to pointing index.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW ARE YOU'.
GOODBYE [Beginner]	Open hand fingers flex open and closed wave.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOODBYE'.
SEE YOU LATER [Beginner]	'V' hand from eye flips forward to 'L' shape.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SEE YOU LATER'.
TAKE CARE [Intermediate]	Two 'K' hands tap crossed at wrists twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TAKE CARE'.

HAVE A NICE DAY [Intermediate]	Cupped hands to chest, 'Good' to open sweeping arc.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HAVE A NICE DAY'.
LONG TIME NO SEE [Intermediate]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
WHAT'S UP [Beginner]	Open 5 hands middle fingers stroke chest upward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHAT'S UP'.
HOW'S EVERYTHING [Intermediate]	Sweeping circle with both 5 hands, thumbs up.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW'S EVERYTHING'.
GLAD TO MEET YOU [Intermediate]	Open palm circular chest rub followed by meet gesture.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GLAD TO MEET YOU'.
FAREWELL [Intermediate]	Formal outward double wave from heart.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FAREWELL'.
CHEERS [Beginner]	Clinking fist motion with open smile.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CHEERS'.
SEE YOU TOMORROW [Intermediate]	'V' hand eye gesture followed by thumb cheek arc.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SEE YOU TOMORROW'.
BLESS YOU [Beginner]	Hands near chin opening forward smoothly.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BLESS YOU'.

■ Section 3: Emergency (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
HELP [Beginner]	Closed fist with thumb upright resting on flat opposite palm.	Lift both hands upward together.	Emergency or assistance request.
EMERGENCY [Beginner]	'E' handshape shaken vigorously side-to-side.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EMERGENCY'.
DANGER [Beginner]	Fist thumbs brushing upwards past opposite knuckles.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DANGER'.
POLICE [Beginner]	'C' handshape tapped twice on left upper chest badge.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'POLICE'.
AMBULANCE [Beginner]	Twisting open 5 hand near head simulating siren light.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'AMBULANCE'.
HOSPITAL [Beginner]	'H' fingers form a cross on opposite shoulder.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOSPITAL'.
DOCTOR [Beginner]	'D' or bent fingertips tapping opposite inner wrist pulse.	Tap wrist pulse twice.	Medical sign for physician or healthcare provider.

FIRE [Beginner]	Wiggling open 5 fingers alternately moving upward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FIRE'.
ACCIDENT [Intermediate]	Two open 'C' hands crash together into 'S' fists.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ACCIDENT'.
PAIN [Beginner]	Index fingers pointing towards each other twisting in opposite directions.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PAIN'.
BLEEDING [Intermediate]	Open hand wiggling downwards from wounded location.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BLEEDING'.
CHOKING [Beginner]	Thumb and index tips touching in circle, other 3 fingers straight.	Hold circular pinch clearly visible.	Sign expressing agreement or confirmation.
ALLERGY [Intermediate]	Index finger touches nose, then moves away dismissively.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ALLERGY'.
MEDICINE [Beginner]	Middle finger grinds in palm of opposite hand.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MEDICINE'.
URGENT [Intermediate]	Quick 'U' hand snaps forward with emphasis.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'URGENT'.
CALL 911 [Beginner]	Phone gesture to ear followed by numbers 9-1-1.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CALL 911'.
CANNOT BREATHE [Intermediate]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
LOST [Beginner]	Fingertips together open and drop downward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LOST'.
TRAPPED [Advanced]	One hand inside clasped fist struggling to move.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TRAPPED'.
SAFETY [Intermediate]	'S' hands crossed at wrists moving outward to protect.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SAFETY'.

■ Section 4: Common Phrases (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
THANK YOU [Beginner]	Flat open hand with fingertips touching chin.	Move fingertips from chin outward toward listener.	Polite expression of gratitude.
PLEASE [Beginner]	Flat open hand rubs center of chest in circular motion.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PLEASE'.
SORRY [Beginner]	'A' fist rubs chest in a circular motion.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SORRY'.

EXCUSE ME [Beginner]	Fingertips brush flat opposite palm forward twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXCUSE ME'.
YES [Beginner]	'S' fist with thumb resting across front fingers.	Nod fist up and down from the wrist like a head nodding.	Affirmative response.
NO [Beginner]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
MAYBE [Beginner]	Flat open palms alternately balance up and down.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MAYBE'.
I AGREE [Intermediate]	Touch forehead index finger, then two index fingers parallel forward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I AGREE'.
I DISAGREE [Intermediate]	Touch forehead, then index fingers pull apart sharply.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I DISAGREE'.
YOU ARE WELCOME [Beginner]	Open hand sweeping inward or nodding with open palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'YOU ARE WELCOME'.
NO PROBLEM [Intermediate]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
I UNDERSTAND [Beginner]	Index finger flicks upward near forehead next to eye.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I UNDERSTAND'.
I DON'T UNDERSTAND [Beginner]	Index finger flicks up with head shaking 'no'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I DON'T UNDERSTAND'.
REPEAT PLEASE [Beginner]	Squished 'O' hand with fingertips bunched together.	Tap lips twice with fingertips.	Sign used to communicate meals or food.
SLOW DOWN [Beginner]	Open flat hand slowly slides up opposite forearm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SLOW DOWN'.
I LIKE IT [Beginner]	Thumb and middle finger pull out from chest pinched.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I LIKE IT'.
I DON'T LIKE IT [Beginner]	Thumb and middle finger pull from chest and flick away.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'I DON'T LIKE IT'.
OF COURSE [Intermediate]	'C' hand circles around resting fist and lands on top.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'OF COURSE'.
I DON'T KNOW [Beginner]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
SOUNDS GOOD [Intermediate]	Point to ear, then sign 'Good' thumbs up.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SOUNDS GOOD'.

■ Section 5: Questions (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
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WHAT [Beginner]	Open palms facing upward shaking side to side gently.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHAT'.
WHERE [Beginner]	Index finger pointing up shaking side to side.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHERE'.
WHEN [Beginner]	Index finger circles and touches tip of opposite index finger.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHEN'.
WHO [Beginner]	Index finger wiggles while thumb rests on chin.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHO'.
WHY [Beginner]	Fingertips touch temple and pull away into 'Y' handshape.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHY'.
HOW [Beginner]	Curved hands back of fingers together roll palms upward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW'.
WHICH [Beginner]	Both 'A' fists thumbs up alternate up and down.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHICH'.
HOW MUCH [Beginner]	Closed fists at waist throw upward into open 5 hands.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW MUCH'.
HOW MANY [Beginner]	Fists facing up open quickly into 5 hands counting.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW MANY'.
ARE YOU READY [Intermediate]	'R' hands shake across body from center outward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ARE YOU READY'.
WHAT HAPPENED [Intermediate]	Both index fingers pointing forward flip palm up.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHAT HAPPENED'.
WHAT TIME [Beginner]	Index finger taps opposite wrist watch twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHAT TIME'.
CAN YOU HELP ME [Beginner]	Closed fist with thumb upright resting on flat opposite palm.	Lift both hands upward together.	Emergency or assistance request.
DO YOU UNDERSTAND [Beginner]	Flick index finger near temple with raised eyebrows.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DO YOU UNDERSTAND'.
WHERE IS THE RESTROOM [Beginner]	'T' handshape shakes side-to-side with question expression.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHERE IS THE RESTROOM'.
CAN I ASK A QUESTION [Intermediate]	Index finger curves into hook moving toward listener.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CAN I ASK A QUESTION'.
WHO IS THAT [Intermediate]	Thumb on chin wiggling index pointing at target.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHO IS THAT'.

WHY NOT [Intermediate]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
HOW OLD ARE YOU [Beginner]	Stroke chin downward like pulling long beard with question face.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOW OLD ARE YOU'.
WHAT DOES THIS MEAN [Intermediate]	'V' fingertips twist on opposite flat palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHAT DOES THIS MEAN'.

■ Section 6: Work & Education (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
STUDENT [Beginner]	Learn sign (take from palm to forehead) + agent marker.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STUDENT'.
TEACHER [Beginner]	Both 'O' hands near head move forward + agent marker.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TEACHER'.
SCHOOL [Beginner]	Clap flat hands horizontally twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SCHOOL'.
COLLEGE [Beginner]	Flat hands together, top hand circles up and settles.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COLLEGE'.
UNIVERSITY [Intermediate]	'U' handshape circles upward off flat opposite palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'UNIVERSITY'.
CLASSROOM [Intermediate]	'C' hands form circle + flat hands outline room walls.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLASSROOM'.
HOMEWORK [Beginner]	Pinch chin-cheek + fist tap on opposite wrist.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOMEWORK'.
EXAM [Intermediate]	Index fingers bend into hooks moving downward in test rows.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXAM'.
ASSIGNMENT [Intermediate]	Hands grasp imaginary paper placing on desk.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ASSIGNMENT'.
PROJECT [Intermediate]	'P' hand moves down 'J' path on opposite palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PROJECT'.
ENGINEERING [Intermediate]	'Y' thumb knuckles link and twist alternately like gears.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ENGINEERING'.
COMPUTER SCIENCE [Advanced]	'C' hand sweeps up forearm + 'Science' pouring beakers.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COMPUTER SCIENCE'.
PRESENTATION [Intermediate]	Flat hands open wide like opening theater curtain.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PRESENTATION'.

MEETING [Beginner]	Both open hands close fingertips together in center.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MEETING'.
DEADLINE [Intermediate]	Flat hand chops straight down on horizontal opposite wrist.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DEADLINE'.
OFFICE [Beginner]	'O' hands outline perimeter of square room.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'OFFICE'.
MANAGER [Intermediate]	'X' hands hold reins shaking + person marker.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MANAGER'.
SCHEDULE [Intermediate]	Four fingers draw grid lines across opposite flat palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SCHEDULE'.
COLLABORATION [Advanced]	Interlocked index/middle fingers swing side to side.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COLLABORATION'.
RESEARCH [Intermediate]	'R' fingertips stroke back of opposite hand repeatedly.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RESEARCH'.

■ Section 7: Health & Medical (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
HEALTHY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'HEALTHY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HEALTHY'.
SICK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SICK'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SICK'.
FEVER [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FEVER'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FEVER'.
COUGH [Beginner]	Standard ASL / ISL manual gesture pattern representing 'COUGH'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COUGH'.
HEADACHE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'HEADACHE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HEADACHE'.
STOMACHACHE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'STOMACHACHE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STOMACHACHE'.
INJURY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'INJURY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'INJURY'.
BANDAGE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'BANDAGE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BANDAGE'.
SURGERY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'SURGERY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SURGERY'.

VACCINE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'VACCINE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'VACCINE'.
NURSE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'NURSE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NURSE'.
CLINIC [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CLINIC'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLINIC'.
PHARMACY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'PHARMACY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PHARMACY'.
PRESCRIPTION [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'PRESCRIPTION'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PRESCRIPTION'.
HEART [Beginner]	Standard ASL / ISL manual gesture pattern representing 'HEART'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HEART'.
BLOOD [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BLOOD'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BLOOD'.
BREATH [Intermediate]	Squished 'O' hand with fingertips bunched together.	Tap lips twice with fingertips.	Sign used to communicate meals or food.
EXAMINE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'EXAMINE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXAMINE'.
RECOVER [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'RECOVER'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RECOVER'.
HEAL [Beginner]	Standard ASL / ISL manual gesture pattern representing 'HEAL'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HEAL'.

■ Section 8: Food & Dining (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
WATER [Beginner]	'W' handshape with three fingers extended.	Tap chin twice with index finger of 'W'.	Sign used to request water.
BREAD [Beginner]	Curved fingers slice down back of opposite curved hand.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BREAD'.
RICE [Beginner]	'R' hand scoops from bowl palm to mouth.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RICE'.
MILK [Beginner]	'S' fist squeezes twice like milking.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MILK'.
TEA [Beginner]	'F' hand dips imaginary tea bag into opposite 'O' cup.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TEA'.

COFFEE [Beginner]	Top fist grinds in circle on lower fist like coffee grinder.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COFFEE'.
APPLE [Beginner]	'X' knuckle twists into cheek.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'APPLE'.
BANANA [Beginner]	Peeing motion of fingers down upright index finger.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BANANA'.
PIZZA [Beginner]	'Z' carved in air with 'V' or 'P' fingers.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PIZZA'.
BURGER [Beginner]	Cupped hands press together and flip like hamburger patty.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BURGER'.
VEGETABLES [Intermediate]	'V' index and middle twist touching cheek twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'VEGETABLES'.
FRUIT [Intermediate]	'F' fingertips twist touching cheek twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FRUIT'.
RESTAURANT [Beginner]	'R' strokes down each side of mouth napkin-style.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RESTAURANT'.
BREAKFAST [Beginner]	Eat + Morning signs linked.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BREAKFAST'.
LUNCH [Beginner]	Eat + Noon signs linked.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LUNCH'.
DINNER [Beginner]	Eat + Night signs linked.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DINNER'.
DELICIOUS [Beginner]	Middle finger snaps off lips outward with smile.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DELICIOUS'.
HUNGRY [Beginner]	'C' hand slides down center of chest to stomach.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HUNGRY'.
THIRSTY [Beginner]	Index finger traces straight down throat.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'THIRSTY'.
SWEET [Beginner]	Fingertips brush downward across chin twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SWEET'.

■ Section 9: Technology & AI (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
COMPUTER [Beginner]	'C' hand arcs bouncing upward along opposite forearm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COMPUTER'.

LAPTOP [Beginner]	Flat hands open hinge-like like opening a notebook laptop.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LAPTOP'.
INTERNET [Beginner]	Middle fingertips touch and twist alternately.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'INTERNET'.
ARTIFICIAL INTELLIGENCE [Intermediate]	Finger spell A-I or 'Smart' + 'Robot/Machine' signs.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ARTIFICIAL INTELLIGENCE'.
MACHINE LEARNING [Advanced]	Gear twists + take knowledge from palm to head.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MACHINE LEARNING'.
SOFTWARE [Intermediate]	'S' hand slides smoothly over flat opposite palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SOFTWARE'.
HARDWARE [Intermediate]	'H' knuckles tap back of solid fist.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HARDWARE'.
MOBILE PHONE [Beginner]	'Y' hand held against ear or tapping flat palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MOBILE PHONE'.
APPLICATION [Intermediate]	'A' hand stamps onto opposite flat palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'APPLICATION'.
DATABASE [Intermediate]	Stacked flat 'C' hands forming multi-tiered cylinder.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DATABASE'.
ALGORITHM [Advanced]	Fingertips trace mathematical formula in air.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ALGORITHM'.
WEBSITE [Beginner]	'W' hand bounces three times horizontally (WWW).	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WEBSITE'.
CAMERA [Beginner]	Index fingers bend in clicking photo shutter motion.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CAMERA'.
MICROPHONE [Beginner]	Fist held at mouth speaking into mic.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MICROPHONE'.
ALGORITHM [Advanced]	Fingers step sequentially across horizontal plane.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ALGORITHM'.
CODE [Intermediate]	'V' fingers cross and uncross like binary logic.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CODE'.
DATA [Intermediate]	Curved fingers stamp rows of coordinates on palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DATA'.
NETWORK [Advanced]	Middle fingers interconnected pulsing outward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NETWORK'.

SERVER [Intermediate]	Stacked horizontal trays moving forward together.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SERVER'.
CYBERSECURITY [Advanced]	Shield hand guarding digital keypad press.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CYBERSECURITY'.

■ Section 10: Numbers & Math (24 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
ZERO [Beginner]	Standard ASL / ISL manual gesture pattern representing 'ZERO'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ZERO'.
ONE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'ONE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ONE'.
TWO [Beginner]	Standard ASL / ISL manual gesture pattern representing 'TWO'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TWO'.
THREE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'THREE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'THREE'.
FOUR [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FOUR'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FOUR'.
FIVE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FIVE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FIVE'.
SIX [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SIX'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SIX'.
SEVEN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SEVEN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SEVEN'.
EIGHT [Beginner]	Standard ASL / ISL manual gesture pattern representing 'EIGHT'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EIGHT'.
NINE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'NINE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NINE'.
TEN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'TEN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TEN'.
TWENTY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'TWENTY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TWENTY'.
FIFTY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FIFTY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FIFTY'.
HUNDRED [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'HUNDRED'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HUNDRED'.

THOUSAND [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'THOUSAND'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'THOUSAND'.
MILLION [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'MILLION'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MILLION'.
PLUS [Beginner]	Standard ASL / ISL manual gesture pattern representing 'PLUS'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PLUS'.
MINUS [Beginner]	Standard ASL / ISL manual gesture pattern representing 'MINUS'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MINUS'.
MULTIPLY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'MULTIPLY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MULTIPLY'.
DIVIDE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'DIVIDE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DIVIDE'.
EQUALS [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'EQUALS'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EQUALS'.
PERCENT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'PERCENT'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PERCENT'.
FRACTION [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'FRACTION'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FRACTION'.
DECIMAL [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'DECIMAL'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DECIMAL'.

■ Section 11: Travel & Transport (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
CAR [Beginner]	Hands grip steering wheel driving forward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CAR'.
BUS [Beginner]	Spell B-U-S or steer large oversized bus wheel.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BUS'.
TRAIN [Beginner]	'H' fingers slide back and forth over opposite 'H' tracks.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TRAIN'.
AIRPLANE [Beginner]	'ILY' handshape glides forward through air like flight.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'AIRPLANE'.
BICYCLE [Beginner]	Fists pedal in alternating circles forward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BICYCLE'.
BOAT [Beginner]	Cupped hands together bobbing on water waves.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BOAT'.

TICKET [Intermediate]	Bent 'V' fingers pinch edge of opposite flat hand.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TICKET'.
PASSPORT [Intermediate]	'P' or stamp hand presses onto flat book palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PASSPORT'.
AIRPORT [Intermediate]	'ILY' plane lands gently on flat runway palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'AIRPORT'.
HOTEL [Intermediate]	'H' flag waving on upright index flagpole.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOTEL'.
MAP [Beginner]	'M' thumbs trace grid boundary lines.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MAP'.
LUGGAGE [Beginner]	Fist holds suitcase handle walking forward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LUGGAGE'.
STATION [Intermediate]	'S' fist lands on flat horizontal platform palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STATION'.
DEPARTURE [Intermediate]	Airplane glides up and away into sky.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DEPARTURE'.
ARRIVAL [Intermediate]	Flat hand swoops down and settles on palm base.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ARRIVAL'.
TRAFFIC [Intermediate]	Open 5 hands alternating past each other like cars.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TRAFFIC'.
ROAD [Beginner]	Parallel flat hands glide forward together straight.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ROAD'.
BRIDGE [Intermediate]	'V' fingers cross under and arch across forearm span.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BRIDGE'.
DESTINATION [Advanced]	Index finger arches through air to land at target spot.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DESTINATION'.
EXPLORE [Intermediate]	Clawed 'C' scans around eyes searching landscape.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXPLORE'.

■ Section 12: Family & People (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
FAMILY [Beginner]	'F' hands touch thumbs, circle around, and meet pinkies.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FAMILY'.
MOTHER [Beginner]	Thumb of open 5 hand taps chin twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MOTHER'.

FATHER [Beginner]	Thumb of open 5 hand taps forehead twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FATHER'.
BROTHER [Beginner]	'L' thumb from forehead drops onto opposite 'L' fist.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BROTHER'.
SISTER [Beginner]	'L' thumb from chin drops onto opposite 'L' fist.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SISTER'.
BABY [Beginner]	Arms cradled rocking imaginary infant back and forth.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BABY'.
CHILD [Beginner]	Flat hand pats top of small child's head twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CHILD'.
FRIEND [Beginner]	Interlocking index finger hooks reversing sides.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FRIEND'.
GRANDMOTHER [Beginner]	Thumb from chin bounces forward in two generational arcs.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GRANDMOTHER'.
GRANDFATHER [Beginner]	Thumb from forehead bounces forward in two generational arcs.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GRANDFATHER'.
SON [Intermediate]	Salute forehead, then cradle baby into crook of arm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SON'.
DAUGHTER [Intermediate]	Salute chin, then cradle baby into crook of arm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DAUGHTER'.
HUSBAND [Intermediate]	C-hand from forehead clasps opposite flat hand.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HUSBAND'.
WIFE [Intermediate]	C-hand from chin clasps opposite flat hand.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WIFE'.
NEIGHBOR [Intermediate]	Near sign + person agent marker.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NEIGHBOR'.
PERSON [Beginner]	Flat hands drop vertically down along body sides.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PERSON'.
PEOPLE [Beginner]	'P' hands cycle alternately forward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PEOPLE'.
MAN [Beginner]	Thumb from forehead touches chest.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MAN'.
WOMAN [Beginner]	Thumb from chin touches chest.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WOMAN'.

COMMUNITY [Intermediate]	Fingertips together twisting outward forming group.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COMMUNITY'.
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■ Section 13: Time & Calendar (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
TIME [Beginner]	Index finger taps opposite wrist watch.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TIME'.
TODAY [Beginner]	'Y' hands bounce downward in front of body twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TODAY'.
TOMORROW [Beginner]	'A' thumb strokes forward off cheek.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TOMORROW'.
YESTERDAY [Beginner]	'S' fist with thumb resting across front fingers.	Nod fist up and down from the wrist like a head nodding.	Affirmative response.
NOW [Beginner]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
LATER [Beginner]	'L' index finger drops downward forward like clock hand.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LATER'.
SOON [Intermediate]	'H' or 'F' thumb and index pinch chin.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SOON'.
ALWAYS [Beginner]	Index finger pointing up circles continuously.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ALWAYS'.
NEVER [Beginner]	Flat hand carves a question-mark style path downward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NEVER'.
DAY [Beginner]	Arm on horizontal table, index arm sets like sun arc.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DAY'.
NIGHT [Beginner]	Bent wrist drops over horizontal arm barrier.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NIGHT'.
WEEK [Beginner]	Index finger slides across opposite flat palm calendar line.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WEEK'.
MONTH [Beginner]	Index finger strokes down back of opposite index.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MONTH'.
YEAR [Beginner]	'S' fist orbits around opposite fist and lands on top.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'YEAR'.
MORNING [Beginner]	Arm lifts up through crook of opposite arm like sunrise.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MORNING'.

AFTERNOON [Beginner]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
EVENING [Beginner]	Curved hand rests over edge of opposite forearm at dusk.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EVENING'.
HOUR [Intermediate]	Index finger makes full 360 clock revolution on palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOUR'.
MINUTE [Intermediate]	Index finger ticks one small second notch on palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MINUTE'.
CALENDAR [Intermediate]	'C' hand tracks down and across open flat book palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CALENDAR'.

■ Section 14: Emotions & States (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
HAPPY [Beginner]	Flat open hands stroke upward on chest repeatedly.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HAPPY'.
SAD [Beginner]	Open 5 hands drop downward in front of drooping face.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SAD'.
EXCITED [Beginner]	Middle fingers alternate brushing upward on chest rapidly.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXCITED'.
ANGRY [Beginner]	Clawed hands pull sharply away from chest/face.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ANGRY'.
CONFUSED [Intermediate]	Point to forehead, then hands twist clawed in opposing circles.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CONFUSED'.
TIRED [Beginner]	Bent flat hands on chest roll downward heavily.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TIRED'.
BORED [Beginner]	Index finger twists at side of nose with sigh.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BORED'.
NERVOUS [Beginner]	Both open hands trembling shaking at waist level.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'NERVOUS'.
CONFIDENT [Intermediate]	'C' hands pull firmly down into solid 'S' fists.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CONFIDENT'.
PROUD [Beginner]	'A' thumb strokes upward up center of chest.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PROUD'.
SCARED [Beginner]	Open 5 hands protectively cross and shake at chest.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SCARED'.

SURPRISED [Beginner]	Index and thumbs pinch at eyes and snap wide open.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SURPRISED'.
PEACEFUL [Intermediate]	Index and middle fingers extended upright in a 'V' shape.	Hold steady in clear victory V.	Sign expressing peace or number two.
FRUSTRATED [Intermediate]	Back of flat hand hits chin or forehead sharply.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FRUSTRATED'.
CURIOUS [Intermediate]	'C' hand gently pinches neck throat area shaking.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CURIOUS'.
HOPEFUL [Intermediate]	Hands near shoulders bend fingers waving down in prayerful wish.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOPEFUL'.
DISAPPOINTED [Intermediate]	Index finger drops down touching chin solemnly.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DISAPPOINTED'.
SHY [Beginner]	Back of curved fingers twist against blushing cheek.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SHY'.
LONELY [Beginner]	Index finger circles alone in front of mouth/chin.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LONELY'.
GRATEFUL [Intermediate]	Both open hands from heart reach gracefully forward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GRATEFUL'.

■ Section 15: Daily Activities (20 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
WAKE UP [Beginner]	Index and thumbs closed at eyes snap open.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WAKE UP'.
SLEEP [Beginner]	Hand over face pulls down into pinched fingers as eyes close.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SLEEP'.
EAT [Beginner]	Squished 'O' hand with fingertips bunched together.	Tap lips twice with fingertips.	Sign used to communicate meals or food.
DRINK [Beginner]	'C' hand tilts toward mouth like drinking from cup.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DRINK'.
COOK [Beginner]	Thumb and index tips touching in circle, other 3 fingers straight.	Hold circular pinch clearly visible.	Sign expressing agreement or confirmation.
CLEAN [Beginner]	Flat hand wipes smoothly across opposite flat palm twice.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLEAN'.
WASH HANDS [Beginner]	Hands rub together vigorously like washing soap.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WASH HANDS'.

SHOWER [Beginner]	Fist above head opens and pulses like water spray.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SHOWER'.
BRUSH TEETH [Beginner]	Index finger scrubs side-to-side across front teeth.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BRUSH TEETH'.
EXERCISE [Beginner]	Fists pump up and down near shoulders like barbell reps.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EXERCISE'.
READ [Beginner]	'V' fingers scan up and down opposite flat palm 'book'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'READ'.
WRITE [Beginner]	Thumb and index pinch scribble across opposite flat palm.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WRITE'.
LISTEN [Beginner]	Cupped hand curved behind ear leaning forward.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LISTEN'.
WATCH [Beginner]	'L' hand near chin points forward focusing eyes.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WATCH'.
DRIVE [Beginner]	Fists grip imaginary steering wheel turning left and right.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DRIVE'.
SHOPPING [Beginner]	Back of hand drops coins out of palm repeatedly.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SHOPPING'.
WALK [Beginner]	Flat hands paddle downward alternately like footsteps.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WALK'.
RUN [Beginner]	'L' hands hook index into opposite thumb and sprint.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RUN'.
STUDY [Beginner]	Fingertips flutter from open book palm towards eyes.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STUDY'.
REST [Beginner]	Arms crossed over chest resting peacefully.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'REST'.

■ Section 16: Actions & Verbs (25 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
GIVE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'GIVE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GIVE'.
TAKE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'TAKE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TAKE'.
BRING [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BRING'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BRING'.

SEND [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SEND'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SEND'.
RECEIVE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'RECEIVE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RECEIVE'.
FIND [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FIND'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FIND'.
LOSE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'LOSE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'LOSE'.
OPEN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'OPEN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'OPEN'.
CLOSE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'CLOSE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLOSE'.
START [Beginner]	Standard ASL / ISL manual gesture pattern representing 'START'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'START'.
STOP [Beginner]	Standard ASL / ISL manual gesture pattern representing 'STOP'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STOP'.
CONTINUE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CONTINUE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CONTINUE'.
WAIT [Beginner]	Standard ASL / ISL manual gesture pattern representing 'WAIT'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WAIT'.
CHANGE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CHANGE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CHANGE'.
BUILD [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BUILD'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BUILD'.
BREAK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BREAK'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BREAK'.
FIX [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FIX'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FIX'.
CREATE [Intermediate]	Squished 'O' hand with fingertips bunched together.	Tap lips twice with fingertips.	Sign used to communicate meals or food.
DESTROY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'DESTROY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DESTROY'.
CHOOSE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'CHOOSE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CHOOSE'.
TRY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'TRY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'TRY'.

WIN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'WIN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WIN'.
SUCCEED [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'SUCCEED'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SUCCEED'.
FAIL [Beginner]	Standard ASL / ISL manual gesture pattern representing 'FAIL'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FAIL'.
GROW [Beginner]	Standard ASL / ISL manual gesture pattern representing 'GROW'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GROW'.

■ Section 17: Colors (15 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
RED [Beginner]	Standard ASL / ISL manual gesture pattern representing 'RED'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RED'.
BLUE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BLUE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BLUE'.
GREEN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'GREEN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GREEN'.
YELLOW [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'YELLOW'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'YELLOW'.
ORANGE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'ORANGE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ORANGE'.
PURPLE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'PURPLE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PURPLE'.
BLACK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BLACK'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BLACK'.
WHITE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'WHITE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WHITE'.
GRAY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'GRAY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GRAY'.
BROWN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BROWN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BROWN'.
PINK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'PINK'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PINK'.
GOLD [Beginner]	Standard ASL / ISL manual gesture pattern representing 'GOLD'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GOLD'.

SILVER [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'SILVER'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SILVER'.
BRIGHT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'BRIGHT'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BRIGHT'.
DARK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'DARK'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'DARK'.

■ Section 18: Places & Nature (24 Signs)

Sign Word	Hand Formation & Finger Mechanics	Motion Trajectory	Context / Purpose
HOUSE [Beginner]	Standard ASL / ISL manual gesture pattern representing 'HOUSE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'HOUSE'.
APARTMENT [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'APARTMENT'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'APARTMENT'.
BUILDING [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'BUILDING'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BUILDING'.
PARK [Beginner]	Standard ASL / ISL manual gesture pattern representing 'PARK'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'PARK'.
BEACH [Beginner]	Standard ASL / ISL manual gesture pattern representing 'BEACH'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'BEACH'.
MOUNTAIN [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'MOUNTAIN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MOUNTAIN'.
FOREST [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'FOREST'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'FOREST'.
RIVER [Beginner]	Standard ASL / ISL manual gesture pattern representing 'RIVER'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RIVER'.
OCEAN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'OCEAN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'OCEAN'.
GARDEN [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'GARDEN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'GARDEN'.
STREET [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'STREET'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STREET'.
CITY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'CITY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CITY'.
COUNTRY [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'COUNTRY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'COUNTRY'.

WORLD [Beginner]	Standard ASL / ISL manual gesture pattern representing 'WORLD'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WORLD'.
SKY [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SKY'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SKY'.
SUN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'SUN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'SUN'.
MOON [Beginner]	Standard ASL / ISL manual gesture pattern representing 'MOON'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'MOON'.
STAR [Beginner]	Standard ASL / ISL manual gesture pattern representing 'STAR'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'STAR'.
RAIN [Beginner]	Standard ASL / ISL manual gesture pattern representing 'RAIN'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'RAIN'.
SNOW [Beginner]	Index and middle fingers snap down onto the thumb.	Quick snapping closure of fingers onto thumb.	Negative response or refusal.
WIND [Beginner]	Standard ASL / ISL manual gesture pattern representing 'WIND'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'WIND'.
CLOUD [Beginner]	Standard ASL / ISL manual gesture pattern representing 'CLOUD'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'CLOUD'.
EARTHQUAKE [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'EARTHQUAKE'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'EARTHQUAKE'.
ISLAND [Intermediate]	Standard ASL / ISL manual gesture pattern representing 'ISLAND'.	Hold gesture steady for 2 seconds to commit text.	Standard sign used in daily communication to express 'ISLAND'.

■■ System Guidelines: Vision AI & 2-Second Hold Protocol

1. 2.0-Second Hold-to-Commit Gate: To prevent false positives or accidental insertions during hand motion, the SignBridge engine evaluates temporal stability. A sign must be held steadily for 2.0 full seconds (2000 ms) with high confidence before it commits into the translated sentence and triggers voice pronunciation.

2. Camera Framing: Position your hand clearly at chest level, approximately 30–50 cm in front of the camera lens with adequate, even lighting.

3. 21 3D Landmark Tracking: The engine runs MediaPipe Hands client-side at 60 FPS, extracting 21 anatomical 3D joints (Wrist, Thumb, Index, Middle, Ring, Pinky) to calculate multi-finger extension ratios, pinch distances, and spatial orientation.

4. Live Sentence Synthesis: Committed words are automatically concatenated, capitalized, punctuated, and vocalized in real-time via the Web Speech API.